

Alina Devkota

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alinadevkota.github.io/

Third-year PhD student in Computer Science at West Virginia University with 4 years of industry experience in Machine Learning and Deep Learning (ML/DL)

Research Interests: Multimodal Learning, Out-of-distribution (OOD) Detection, Foundation Models, Federated Learning, Vision Language Models (VLM), Large Language Models (LLM), Medical Imaging

EDUCATION

Lane Department of Computer Science and Electrical Engineering

West Virginia University, Morgantown, WV, USA

Anticipated Graduation: Dec. 2027

PhD in Computer Science

Department of Electronics and Computer Engineering

Tribhuvan University, Nepal

September 2019

Bachelor's in Computer Engineering

PUBLICATIONS (Full Paper List: <https://scholar.google.com/citations?user=oo43-CUAAAAJ&hl=en&oi=ao>)

- Devkota A., et al. *FedVG: Gradient-Guided Aggregation for Enhanced Federated Learning*. **Findings of the Conference on Computer Vision and Pattern Recognition (CVPR), 2026**
- Devkota A., et al. *AI analysis for ejection fraction estimation from 12-lead ECG*. **Scientific Reports (Nature Portfolio), 2025**
- Thrasher J., Devkota A., et al. *Multimodal federated learning in healthcare: a review*. **Journal of Healthcare Informatics Research, 2025**
- Thrasher J., Devkota A., et al. *Te-SSL: Time and event-aware self-supervised learning for Alzheimer's disease progression analysis*. **MICCAI, 2024**
- Mu, James, et al. *Towards Federated Learning Across Biobanks: Prototype Software*. **Carnegie Mellon University–NVIDIA Hackathon, 2026**
- Amgain, Sanskar, et al. *Local K-Similarity Constraint for Federated Learning with Label Noise*. **International Symposium on Biomedical Imaging (ISBI), 2025**

PUBLICATIONS UNDER REVIEW

- Devkota A., et al. *Federated Foundation Model for GI Endoscopy Images*. **Npj Digital Medicine, 2026**

SELECTED PROJECTS

Multimodal Out-of-Distribution (OOD) Detection

Dec. 2025 – Present

- Developing methods to detect out-of-distribution inputs across multiple data modalities, primarily focusing on Vision Language Models to improve AI robustness and reliability

Jailbreaking Multimodal Unlearning Models

Jan. 2026 – Present

- Investigating jailbreak and adversarial prompting strategies against multimodal unlearning models to evaluate unlearning reliability and identify failure modes under malicious or adaptive attacks

Rationale-Augmented Knowledge Distillation for Small Language Models

Sep. 2025 – Dec. 2025

- Developed a multi-task knowledge distillation framework to train compact language models (e.g., T5-Small, FLAN-T5-Small) to jointly predict task labels and reconstruct LLM-generated rationales

Feature Selection in Manufacturing using Knockoff Framework

Dec. 2024 – Present

- Implemented the knockoff framework for feature selection in manufacturing to select 44 relevant features out of 590 while controlling false discovery rates

Contrastive Learning Framework for Knowledge Distillation

April 2024 - Jan. 2025

- Conducted robustness analysis against adversarial attacks and imbalanced data

AWARDS AND ACADEMIC SERVICES

- **Web and Publicity Chair**, *4th DEMI Workshop*, MICCAI 2026
- **Second Place**, *Rapid Fire Presentation*, Statler College Research Week, West Virginia University, 2026
- **Reviewer**, *DEMI 2025 (MICCAI Workshop) and MICCAI 2026*
- **Lecturer**, *WVU SURE Generative AI Workshop*, West Virginia University, 2025
- **Second Place**, *Poster Presentation*, *WVU AI Symposium*, West Virginia University, 2025
- **Second Place and Honorable Mention**, *Graduate Poster Competition*, West Virginia University, 2024
- **Presenter**, *Global AI Bootcamp*, Nepal Cloud Professionals, 2023
- **Teaching Assistant**, *Third Nepal Winter School in AI*, NAAMII, 2021
- **Speaker**, *Part of Research Training*, Fusemachines Nepal, 2021
- **Best Project Award**, *Department of Computer Science and Electrical Engineering*, Tribhuvan University, 2019
- **Fusemachines AI Fellowship**, Fusemachines Nepal, 2019
- **Winner**, *at App-a-thon*, Spiralogs International, 2017
- **Academic Excellence Scholarship**, Tribhuvan University, 2015-2019

LEADERSHIP EXPERIENCE

- **Organizer**, *LOCUS*, Tribhuvan University, 2018 - 2019
- **Volunteer**, *Ministry of Education, Nepal*, in collaboration with NSDEVIL, 2019
- **Trainer**, *CAN Cybersecurity Taskforce*, 2018
- **Trainer**, *Children in Technology*, 2017
- **Department Coordinator**, *Kathmandu Valley Leo Club, District 325 A1*, 2015 - 2016

EXPERIENCE

West Virginia University, Morgantown, WV

Jan. 2024 – Present

Graduate Research Assistant

- Built and evaluated a diverse set of deep learning and classical models on a 55K-patient ECG cohort to estimate left ventricular ejection fraction in a rural Appalachian population
- Trained a foundation model for GI endoscopy images using federated learning, improving disease classification accuracy by 8.9% while preserving data privacy
- Developed FedVG, a gradient-guided federated aggregation method that adaptively weights clients using global validation gradients, improving generalization across heterogeneous client distribution

coac GmbH, Cologne, Germany

April 2022 – Dec. 2023

Machine Learning Engineer

- Enhanced OCR and object detection in PDFs containing scanned schematic diagrams, improving recognition accuracy and inference speed
- Investigated optimization algorithms to minimize pandemic lockdowns in German counties, and used AI models to replicate a mathematical model to reduce computation time from ~5 hours to ~1 minute with an acceptable tolerance of 5%

NepAI Applied Mathematics and Informatics Institute for Research (NAAMII), Nepal July 2021 – July 2022

Research Assistant, *Part-time*

- Applied UNET for fetal head segmentation to measure head circumference (HC18 dataset)
- Used a classification algorithm to detect cancer in ultrasound images of breasts

Fusemachines, Nepal

Sep. 2019 – April 2021

Machine Learning Engineer

- Led ML pipeline development to reconstruct 3D bones from X-rays (Intelligent surgery project)
- Implemented pseudolabelling to develop a student status engine (FuseClassroom, an online LMS)
- Integrated Elasticsearch for plagiarism detection, significantly reducing the search space and computation time in large-scale text comparisons

SKILLS

- **Programming:** Python, C, C++, R, JavaScript, MATLAB
- **Machine Learning & Deep Learning:** PyTorch, Scikit-learn, OpenCV, Huggingface, Transformers, Multimodal Learning, Foundation Models, Federated Learning, VLM, LLM, TensorFlow, LangChain
- **Data Analysis & Visualization:** NumPy, Pandas, Matplotlib
- **Software & Tools:** Slurm, Git/GitHub, Ray, ClickUp, Jira, Docker, AWS, FastAPI, Django, REST APIs, SQL, MongoDB